Description

Image





Caption

Ductile or malleable cast irons are used for heavily loaded parts such as gears and automotive suspension components.

The material

The foundations of modern industrial society are set, so to speak, in cast iron: it is the material that made the industrial revolution possible. Today it holds a second honor: that of being the cheapest of all engineering metals. Cast iron contains at least 2% carbon -- most have 3 to 4% -- and from 1-3% silicon. The carbon makes the iron very fluid when molten, allowing it to be cast to intricate shapes. There are five classes of cast iron: gray, white, ductile (or nodular), malleable and alloy; details are given under Design Guidelines, below. The two that are most used are gray and ductile. This record is for ductile cast iron.

Compositional summary

Fe/3.2-4.1%C/1.8-2.8%Si/<0.8%Mn/<0.1%P/<0.03%S

General properties

Density		7.05e3	-	7.15e3	kg/m^3
Price	*	0.296	-	0.32	GBP/kg
Date first used		1948			

Mechanical properties

Young's modulus		170	-	180	GPa
Shear modulus		64	-	71	GPa
Bulk modulus		120	-	140	GPa
Poisson's ratio		0.27	-	0.28	
Yield strength (elastic limit)		246	-	630	MPa
Tensile strength		400	-	900	MPa
Compressive strength	*	273	-	639	MPa

Elongation		2	-	26	% strain
Hardness - Vickers		135	-	290	HV
Fatigue strength at 10^7 cycles		195	-	317	MPa
Fracture toughness		23.3	-	51.6	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	6e-4	-	0.0018	

Thermal properties

Melting point		1.13e3	-	1.25e3	°C
Maximum service temperature		350	-	450	°C
Minimum service temperature	*	-69.2	-	-0.15	°C
Thermal conductor or insulator?		Good conductor			
Thermal conductivity	*	31	-	42	W/m.°C
Specific heat capacity		460	-	495	J/kg.°C
Thermal expansion coefficient		10	-	12.5	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good conductor			
Electrical resistivity		49	-	56	µohm.cm

Optical properties

Transparency		Opaque			

Critical Materials Risk

High critical material risk?		No			

Processability

Castability		5			
Formability		1	-	2	
Machinability		4			
Weldability		3	-	4	
Solder/brazability		1	-	2	

Durability: water and aqueous solutions

Water (fresh)		Acceptable			
Water (salt)		Limited use			
Soils, acidic (peat)		Acceptable			
Soils, alkaline (clay)		Acceptable			
Wine		Unacceptable			

Durability: acids

Acetic acid (10%)	Limited use
Acetic acid (glacial)	Limited use
Citric acid (10%)	Limited use
Hydrochloric acid (10%)	Unacceptable
Hydrochloric acid (36%)	Unacceptable
Hydrofluoric acid (40%)	Unacceptable
Nitric acid (10%)	Unacceptable
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Unacceptable
Phosphoric acid (85%)	Unacceptable
Sulfuric acid (10%)	Acceptable
Sulfuric acid (70%)	Acceptable

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Acceptable

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Excellent
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Limited use
Acetone	Excellent
Ethyl alcohol (ethanol)	Acceptable
Ethylene glycol	Acceptable
Formaldehyde (40%)	Limited use

Glycerol	Excellent
Methyl alcohol (methanol)	Acceptable

Durability: halogens and gases

Chlorine gas (dry)	Excellent
Fluorine (gas)	Limited use
O2 (oxygen gas)	Limited use
Sulfur dioxide (gas)	Acceptable

Durability: built environments

Industrial atmosphere	Acceptable
Rural atmosphere	Acceptable
Marine atmosphere	Limited use
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Excellent
Tolerance up to 450 C (842 F)	Excellent
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	2.23e9	tonne/yr
Reserves, principal component	1.59e11	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 2.24	-	2.46	kg/kg
Embodied energy (virgin grade production)	* 30.4	-	33.6	MJ/kg
Water usage	* 42.6	-	47.1	l/kg

Material processing: energy

Metal casting, embodied energy	* 45.4	-	52.1	MJ/kg
Chemical vapor deposition (CVD), embodied energy	* 1e4	-	1.13e4	MJ/kg
Machining, coarse, embodied energy	* 17.7	-	19.9	MJ/kg
Machining, fine, embodied energy	* 24	-	27.4	MJ/kg

Machining, grinding, embodied energy	* 30.3	-	35	MJ/kg
Machining, laser, CO2, embodied energy	* 1.19	-	1.34	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 3.24	-	3.75	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 823	-	926	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.16	-	1.3	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.57	-	1.8	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.98	-	2.29	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.098	-	0.11	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 0.493	-	0.541	kg/kg
Embodied energy, recycling	* 9.12	-	10.1	MJ/kg
Recycle fraction in current supply	50			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

As metals go, it takes relatively little energy to make cast iron; it's exceptionally durable, and easily recycled. The pollution caused by blast furnaces in which it is made was at one time a major problem; but modern technology has totally overcome this.

Supporting information

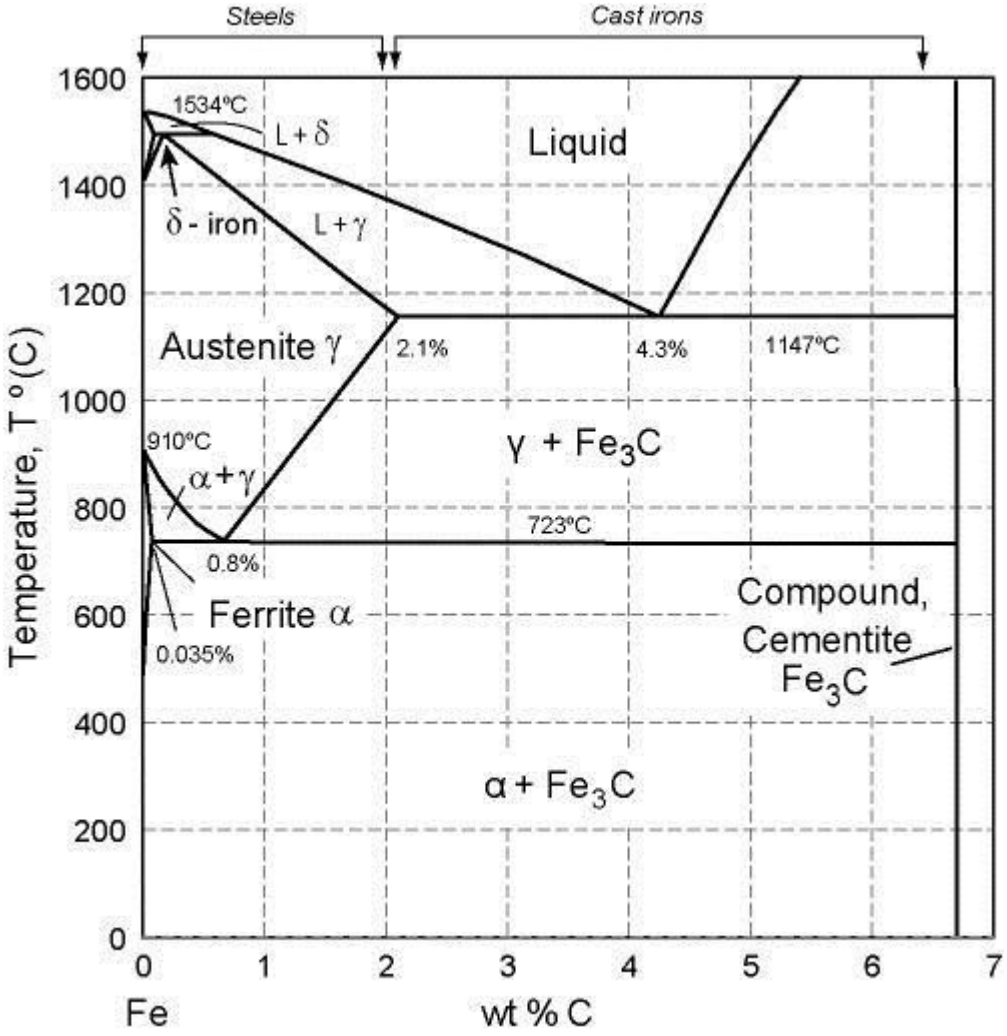
Design guidelines

There are five kinds of cast iron. Gray cast iron machines easily, damps vibration well, is relatively brittle and has low tensile strength; it is the material of automotive cylinder blocks, exhaust manifolds, break disks and drums, gears and flywheels. White cast iron, made by chill casting to give a high cooling rate at the surface, is much harder than gray; it is used when wear resistance is wanted, as in rolls for rolling mills, blades for crushers and mixers. Nodular (ductile) cast iron contains additions that cause the flakes of graphite that are present in gray iron to spherodize, giving higher toughness and strength but at the loss of damping-ability; it is used for crank shafts and heavy duty gears. Malleable cast iron, made by heat-treating white cast iron, is ductile and easily machined; it is used for heavy-duty parts of cars, trucks, and railway rolling stock. Finally, alloy cast irons contain up to 35% of chromium or nickel; they are corrosion resistant and have high strength, but are much more expensive.

Technical notes

There is no single systematic numbering system for cast irons. The UNS and the AISI systems are widely used, particularly in the US. Ductile cast irons are generally described using the ASTM system, a set of three numbers separated by colons, like this: ASTM 60:40:18. The first number is the tensile strength in ksi, the second is the yield strength in ksi and the third is the elongation in % (to convert ksi to MPa, multiply by 6.89).

Phase diagram



Phase diagram description

Grey cast irons are based on iron (Fe) with 3 - 4.1% carbon (C), for which this is the phase diagram. Ductile cast irons have small additions of magnesium or cerium that cause the graphite to form spherical nodules rather than flakes.

Typical uses

Brake discs and drums, bearings, camshafts, cylinder liners, piston rings, machine tool structural parts, engine blocks, gears, crankshafts, heavy-duty gear cases, pipe joints, pump casings, components in rock crushers.

Links

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Producers

Reference

Values marked * are estimates.
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